

Problem 1. Regularized MMSE Criterion

A variation of the MMSE criterion minimizes the norm of the weight vector $\mathbf{w}$. This is a way to control the complexity of the structure. The corresponding function is

$$\mathcal{L}(\mathbf{x}, \mathbf{w}) = \mathbb{E}[e^2] + \lambda \|\mathbf{w}\|^2$$

- (1) Make the derivation of the closed solution for $\mathbf{w}$.
- (2) Work out an iterative solution using the same technique as used in the Least Mean Squares algorithm.
- (3) Comment and compare both solutions in a short conclusion section.

Solution.

We are given the regularized MMSE cost

$$J(\mathbf{w}) = \mathbb{E}[(d - \mathbf{w}^\top \mathbf{x})^2] + \lambda \|\mathbf{w}\|^2,$$

where $\mathbf{x}$ is the input vector, d is the desired response, $\mathbf{w}$ is the weight vector we want to design, and $\lambda \geq 0$ is the regularization parameter that keeps the weights from growing too large.

(1) Closed-form solution

Start from the MSE part:

$$\mathbb{E}[(d - \mathbf{w}^\top \mathbf{x})^2] = \mathbb{E}[d^2] - 2\mathbf{w}^\top \mathbb{E}[d\mathbf{x}] + \mathbf{w}^\top \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \mathbf{w}.$$

To simplify notation, let

$$\mathbf{p} = \mathbb{E}[d\mathbf{x}] \quad \text{and} \quad \mathbf{R} = \mathbb{E}[\mathbf{x}\mathbf{x}^\top].$$

Then the cost becomes

$$J(\mathbf{w}) = \mathbb{E}[d^2] - 2\mathbf{p}^\top \mathbf{w} + \mathbf{w}^\top \mathbf{R} \mathbf{w} + \lambda \mathbf{w}^\top \mathbf{w}.$$

Combine the quadratic terms:

$$J(\mathbf{w}) = \mathbb{E}[d^2] - 2\mathbf{p}^\top \mathbf{w} + \mathbf{w}^\top (\mathbf{R} + \lambda \mathbf{I}) \mathbf{w}.$$

Now differentiate w.r.t. $\mathbf{w}$:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -2\mathbf{p} + 2(\mathbf{R} + \lambda \mathbf{I}) \mathbf{w}.$$

Set this to zero to find the minimizer:

$$(\mathbf{R} + \lambda \mathbf{I}) \mathbf{w} = \mathbf{p}.$$

Assuming $\mathbf{R} + \lambda \mathbf{I}$ is invertible (it is if $\lambda > 0$), we get

$$\boxed{\mathbf{w}^* = (\mathbf{R} + \lambda \mathbf{I})^{-1} \mathbf{p}.}$$

(2) Iterative solution (LMS-style)

For an online / adaptive version, we replace the expectations with the current sample $(\mathbf{x}(n), d(n))$. Define the instantaneous error

$$e(n) = d(n) - \mathbf{w}^\top(n) \mathbf{x}(n).$$

The instantaneous cost is

$$J_n(\mathbf{w}) = e^2(n) + \lambda \|\mathbf{w}\|^2.$$

Compute its gradient:

$$\nabla_{\mathbf{w}} J_n(\mathbf{w}) = -2e(n)\mathbf{x}(n) + 2\lambda\mathbf{w}(n).$$

A standard stochastic-gradient update with step size μ is

$$\begin{aligned}\mathbf{w}(n+1) &= \mathbf{w}(n) - \frac{\mu}{2} \nabla_{\mathbf{w}} J_n(\mathbf{w}(n)) \\ &= \mathbf{w}(n) + \mu e(n)\mathbf{x}(n) - \mu\lambda\mathbf{w}(n).\end{aligned}$$

So the final iterative rule is

$$\boxed{\mathbf{w}(n+1) = (1 - \mu\lambda) \mathbf{w}(n) + \mu e(n)\mathbf{x}(n).}$$

This is exactly LMS but with a “weight decay” factor $(1 - \mu\lambda)$.

(3) Short comparison

- *Batch / closed form:* $\mathbf{w}^* = (\mathbf{R} + \lambda\mathbf{I})^{-1}\mathbf{p}$ is precise but needs $\mathbf{R}$ and $\mathbf{p}$ and a matrix inverse.
- *Iterative / LMS-like:* $\mathbf{w}(n+1) = (1 - \mu\lambda)\mathbf{w}(n) + \mu e(n)\mathbf{x}(n)$ works sample by sample and is good when data arrive over time.
- *Effect of λ :* In both cases λ penalizes large weights, makes the problem better conditioned, and usually improves generalization.